

Physical data model

Core Tables

Article

Column Name	Description	SQL Type	Constraints
ArticleID	Unique article identifier	SERIAL	Primary key
Title	Article title	VARCHAR(500)	NOT NULL
Abstract	Article summary	TEXT	NOT NULL
PublicationDate	Publication date	DATE	NOT NULL
DOI	Digital Object Identifier	VARCHAR(255)	UNIQUE, NOT NULL
JournalID	Journal reference	INT	Foreign key to Journal

Author

Column Name	Description	SQL Type	Constraints
AuthorID	Unique author identifier	SERIAL	Primary key
FirstName	Author's first name	VARCHAR(100)	NOT NULL
LastName	Author's last name	VARCHAR(100)	NOT NULL
ORCID	ORCID identifier	VARCHAR(20)	UNIQUE
Email	Contact email	VARCHAR(255)	NOT NULL

Journal

Column Name	Description	SQL Type	Constraints
JournalID	Unique journal identifier	SERIAL	Primary key
Name	Journal name	VARCHAR(255)	NOT NULL
ISSN	ISSN number	CHAR(9)	UNIQUE, NOT NULL
Publisher	Publisher name	VARCHAR(255)	NOT NULL

Institution

Column Name	Description	SQL Type	Constraints
InstitutionID	Unique institution identifier	SERIAL	Primary key
Name	Institution name	VARCHAR(255)	NOT NULL
Address	Physical address	TEXT	NOT NULL

Keyword

Column Name	Description	SQL Type	Constraints
KeywordID	Unique keyword identifier	SERIAL	Primary key
Term	Keyword term	VARCHAR(100)	UNIQUE, NOT NULL

Junction Tables

ArticleAuthor

Column Name	Description	SQL Type	Constraints
ArticleID	Article reference	INT	Foreign key, Composite PK
AuthorID	Author reference	INT	Foreign key, Composite PK

ArticleKeyword

Column Name	Description	SQL Type	Constraints
ArticleID	Article reference	INT	Foreign key, Composite PK
KeywordID	Keyword reference	INT	Foreign key, Composite PK

Relationship Tables

Affiliation

Column Name	Description	SQL Type	Constraints
AuthorID	Author reference	INT	Foreign key, Composite PK
InstitutionID	Institution reference	INT	Foreign key, Composite PK
StartDate	Affiliation start	DATE	NOT NULL
EndDate	Affiliation end	DATE	Optional

Article_history

Column Name	Description	SQL Type	Constraints
ArticleID	Article reference	INT	Foreign key
ValidFrom	Validity start date	DATE	NOT NULL
ValidTo	Validity end date	DATE	NOT NULL